

Sunpill Kim

Curriculum Vitae

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RESEARCH FOCUS

My research focuses on the security and vulnerability analysis of metric-learning-based recognition systems, including biometric and vision-language models. My work spans adversarial attacks, biometric template protection, and related detection tasks, with an emphasis on both theoretical foundations and practical deployment.

WORK EXPERIENCE

Sookmyung Women's University, Seoul, South Korea

- Assistant Professor, Division of Software Sep 2026 - Present

Hanyang University, Seoul, South Korea

- Postdoctoral Researcher, Research Institute for Natural Sciences Mar 2026 - Aug 2026
- Advisors: Jae Hong Seo

Institute for Infocomm Research (I²R), A*STAR, Singapore

- Ph.D. Researcher (ARAP Scholar), Cybersecurity Department Jan 2023 - Jan 2024
- Supervisors: Dr. Khin Mi Mi Aung and Dr. Yong Kiam Tan

EDUCATION

Hanyang University, Seoul, South Korea

- Ph.D. in Mathematics (Applied Mathematics) Mar 2020 - Feb 2026
 - Thesis: Score-Based Non-Adaptive Attack Against Face Recognition Systems
 - Advisor: Jae Hong Seo
- B.S. in Mathematics Mar 2015 - Feb 2020

PUBLICATIONS

†: Equally contributed.

CONFERENCE

- [C9] “Casting the Net! Revisiting MasterFace Impersonation Attacks”
Seunghun Paik[†], Sunpill Kim[†], Chanwoo Hwang, Jae Hong Seo
ACM CCS 2026 (acceptance rate: 15.8%)
- [C8] “Non-Adaptive Adversarial Face Generation”
Sunpill Kim, Seunghun Paik, Chanwoo Hwang, Minsu Kim, Jae Hong Seo
NeurIPS 2025 (acceptance rate: 24.5%)
- [C7] “IDFace: Efficient and Secure Identification for Face Images”
Sunpill Kim[†], Seunghun Paik[†], Chanwoo Hwang, Dongsu Kim, Junbum Shin, Jae Hong Seo
ICCV 2025 (acceptance rate: 24.0%)
- [C6] “A Survey of Model Inversion Attacks on Image Domain”
Changjin Kim, Chanwoo Hwang, Sunpill Kim, Jae Hong Seo
IEEE ICTC 2025
- [C5] “Towards Certifiably Robust Face Recognition”
Seunghun Paik, Dongsu Kim, Chanwoo Hwang, Sunpill Kim, Jae Hong Seo
ECCV 2024
- [C4] “On the Certifiable Robustness of Face Recognition Systems”
Seunghun Paik, Dongsu Kim, Chanwoo Hwang, Sunpill Kim, Jae Hong Seo
CISC-S 2024 (South Korea)
- [C3] “Scores Tell Everything about Bob: Non-adaptive Face Reconstruction on Face Recognition Systems”

Sunpill Kim, Yong Kiam Tan, Bora Jeong, Soumik Mondal, Khin Mi Mi Aung, Jae Hong Seo
IEEE S&P 2024 (acceptance rate: 17.8%)

- [C2] “Security Analysis on Locality-Sensitive Hashing-based Biometric Template Protection Schemes”
Seunghun Paik, Sunpill Kim, Jae Hong Seo
BMVC 2023
- [C1] “IronMask: Modular Architecture for Protecting Deep Face Template”
Sunpill Kim, Yunseong Jeong, Jinsu Kim, Jungkon Kim, Hyung Tae Lee, Jae Hong Seo
CVPR 2021 (acceptance rate: 23.4%)

JOURNAL

- [J6] “SilverMask: Face Template Protection with Fine-Grained Noise-Correction”
Minsu Kim[†], Seunghun Paik[†], Seongae Baek, Sangyoon Shin, Sunpill Kim, and Jae Hong Seo
IEEE Access, 2026, doi: 10.1109/ACCESS.2026.3689766
- [J5] “Doubly Efficient Fuzzy Private Set Intersection for High-dimensional Data with Cosine Similarity”
Hyunjung Son, Seunghun Paik, Yunki Kim, Sunpill Kim, Jae Hong Seo
IEEE Access, 2025, doi: 10.1109/ACCESS.2025.3648455.
- [J4] “Towards Certifiably Robust Face Recognition: Analyses and Improvements”
Seunghun Paik, Dongsoo Kim, Chanwoo Hwang, Sunpill Kim, Jae Hong Seo
IEEE Transactions on Biometrics, Behavior, and Identity Science, 2025, doi: 10.1109/TBIOM.2025.3644396.
- [J3] “On the Reversibility of Locality-Sensitive Hashing-based Biometric Template Protections”
Seunghun Paik, Chanwoo Hwang, Sunpill Kim, Jae Hong Seo
IEEE Transactions on Dependable and Secure Computing, 2025, doi: 10.1109/TDSC.2025.3637307.
- [J2] “Deep Face Template Protection in the Wild”
Sunpill Kim, Hoyong Shin, Jae Hong Seo
Pattern Recognition, 2025, doi: 10.1016/j.patcog.2024.111336.
- [J1] “Analysis on Secure Triplet Loss”
Bora Jeong, Sunpill Kim, Seunghun Paik, Jae Hong Seo
IEEE Access, 2022, doi: 10.1109/ACCESS.2022.3225430.

PREPRINTS

- [P3] “Naïve Exposure of Generative AI Capabilities Undermines Deepfake Detection”
Sunpill Kim[†], Chanwoo Hwang[†], Minsu Kim, Jae Hong Seo
arXiv, Available at <https://arxiv.org/abs/2603.10504>
- [P2] “Scores Know Bob’s Voice: Speaker Impersonation Attack”
Chanwoo Hwang, Sunpill Kim, Yong Kiam Tan, Tianchi Liu, Seunghun Paik, Dongsoo Kim, Mondal Soumik, Khin Mi Mi Aung, Jae Hong Seo
arXiv, Available at <https://arxiv.org/abs/2603.02781>
- [P1] “Formalization of the Schwartz-Zippel Lemma”
Sunpill Kim[†] and Yong Kiam Tan[†]
Archive of Formal Proofs, 2023

HONORS & AWARDS

AI Seoul Tech Fellowship , Seoul Future Foundation	Jun 2026
– Selected as representative awardee speaker for achieving the top evaluation result; postdoctoral research grant (\$44K).	
The Outstanding Ph.D Dissertation Award of University , Hanyang University	Feb 2026
The 1st Graduate Presidential Science Scholarship , Korea Student Aid Foundation	2024 - 2026
– Sole recipient in Mathematics Ph.D. program; full living support (\$24K/year).	
<i>Best Award</i> , Best Research Paper Award 2024, The Research Institute for Natural Sciences, Hanyang University	2025
A*STAR Research Attachment (ARAP) , Agency for Science, Technology and Research, Singapore	2023 - 2024
– Fully funded research attachment (≈\$47K).	
<i>Excellence Award</i> , National Cryptographic Technology Contest.	2023
<i>The Samil Scholarship</i> , The Samil Foundation.	2022 - 2023

RESEARCH PROJECT

Secure Authentication System using Deep Learning-based Biometric Recognition System, NRF (PI)	2024 - 2025
Computer-Aided Cryptography for Zero-Knowledge Proofs and Verifiable Computing, A*STAR (Participant)	2023 - 2024

PATENT

- [P4] Similarity-Private Set Intersection Protocol Using Cosine Similarity-based Similarity Measurement between High-dimensional Data
Jae Hong Seo, Hyeonjeong Son, Seunghun Paik, Yunki Kim, Sunpill Kim, Dongwoo Kim, Heewon Chung
KOR 10-2025-0200142
- [P3] Protocol System for Real-valued Error Correcting Code using Commutative Algebraic Structure over Hypersphere
Jae Hong Seo, Sunpill Kim, Sangyun Shin, Sungae Baik, Minsu Kim, Seunghun Paik
KOR 10-2025-0008685 (10-2941630)
- [P2] Server and method for identifying target user thereof
Sunpill Kim, Seunghun Paik, Chanwoo Hwang, Dongsu Kim, Jae Hong Seo, Junbum Shin, Jung Woo Kim
KOR 10-2024-0031957 and USPTO 18/598,233 (12,476,815)
- [P1] Protocol System for Real-valued Error Correcting Code over Hypersphere
Jae Hong Seo, Sunpill Kim, Sangyun Shin, Sungae Baik, Minsu Kim, Seunghun Paik
KOR 10-2023-0178374 (10-2941634)

PROFESSIONAL SERVICES

REVIEWER

Journals: IEEE Transactions on Information Forensics and Security, IEEE Transactions on Dependable and Secure Computing
Conferences: NeurIPS 2026, ECCV 2026, CVPR 2026, CVPR 2025, BMVC 2024, CVPR 2024, PKC 2023, ASIACRYPT 2021, ProvSec 2020

TEACHING EXPERIENCE

Sookmyung Women's University, Seoul, South Korea

- Mathematics for Computer Science; Education of Artificial Intelligence; Advanced Engineering Mentoring Fall 2026

Hanyang University, Seoul, South Korea

- Mathematical Algorithms (Teaching Fellow) Spring 2025
- Number Theory; Capstone PBL (Teaching Assistant) 2020 - 2021

INVITED TALKS

Hanyang University, Department of Mathematics "Non-Adaptive Adversarial Face Generation"	Oct 2025
Hanyang University, Department of Mathematics "Are Deep-Learning Based Face Recognition Systems Secure?"	May 2024
Desilo Inc. (Industry Talk) "Biometric Information Extraction Threats and Countermeasures in Deep Learning-based Face Recognition System"	Dec 2022
Korean Artificial Intelligence Association & LG AI Research "IronMask: Modular Architecture for Protecting Deep Face Template"	Nov 2021